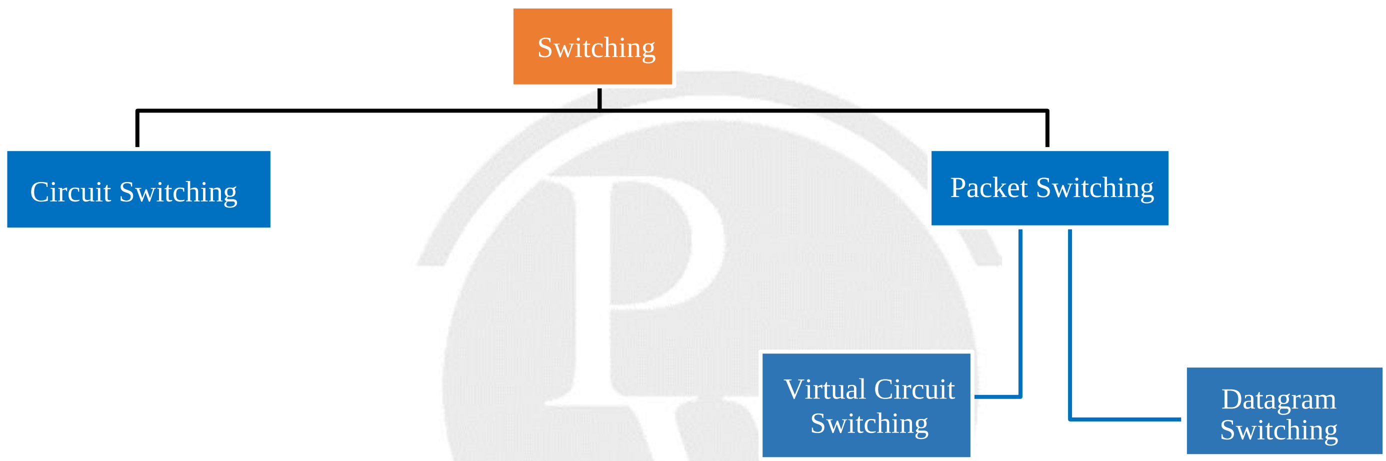
7 ROUTING ALGORITHMS, SWITCHING & IP SUPPORT PROTOCOL

7.1 Switching



Circuit Switching	Packet switching
(I) It has three phases-connection establishment, Data transfer and Connection termination.	It has only one phase-Data transfer
(2) Physical path between source and destination	No physical path
(3) All packets use same path	Packet may follow different path (travel independently)
(4) Reserves the entire bandwidth in advance	Does not reserve
(5) Bandwidth wastage	No Bandwidth wastage
(6) No store and forward transmission	Supports store and forward transmission
(7) Congestion can happen during connection establishment phase	Congestion can happen during data transfer phase

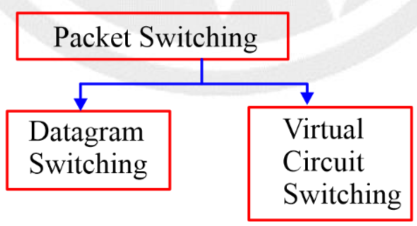
(8) It is reliable	Not reliable
(9) Better for sending large messages	Better for sending small messages
(10) Not fault tolerant technique	Fault tolerant technique
(11) Circuit switching is implemented at physical layer.	Packet switching is implemented at network layer
Total time= Setup time $+ T_d + P_d +$ Tear down time $TT = S + \frac{L}{B} + X.\frac{d}{V} + T$	For X Hop and N packet Total time= $X\,[T_d + P_d] + X - 1[P_{rd} + Q_d] + N - 1(T_d)$

7.2 Generalized Formula for optimal packet size(P)

M = Message size
h = Header size
p = Payload/Packet data size
No. of Hops = X

$$p = \sqrt{\frac{Mh}{X-1}}$$

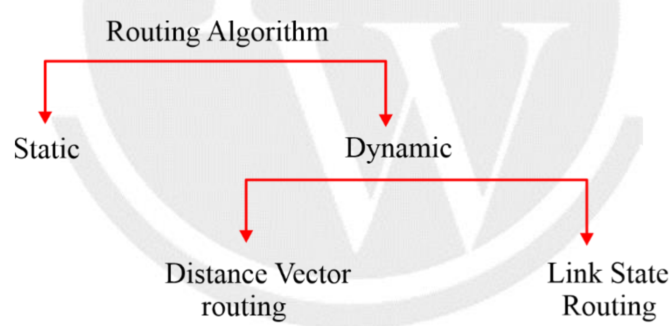
So optimum packet size P = p + h



Datagram Switching	Virtual circuit switching
(1) It is a connection less service.	It is connection-oriented service.
(2) All the packets may follow different path.	All the packet follows the same dedicated path.
(3) Data may appear out of order at the destination since the packets may follow different paths.	Data appears in order at the destination since all the packets take the same dedicated path.
(4) It is not reliable since packets may be discarded.	It is highly reliable.

(5) Cheap	Costly
(6) No resource reservation is done.	First packet reserves the resources (CPU, bandwidth and buffer) for the subsequent packets.
(7) All the packets require a global header which contains full information about the destination.	Only first packet requires a global header which identifies the path from one end to another end. All the following packets require a local header which identifies the path from hop to hop.
(8) The packets may be discarded at intermediate switches if sufficient resources are not available to process the packets.	The packets are never discarded at intermediate switches and immediately forwarded since resources are reserved for them.
(9) IP Networks uses datagram switching.	ATM (Asynchronous Transfer Mode) uses virtual circuit switching.
(10) Datagram switching is normally implemented at network layer.	Virtual circuit switching is normally implemented at data link layer.

7.3 Routing Algorithm



Distance vector Routing	Link state Routing
1. 1980’s	1. 1990’s
2. Bandwidth required is very less because we sent only distance vector packet.	2. Band width required is high because we sent entire link state packet
3. Local knowledge	3. Global knowledge
4. Bellman Ford algorithm	4. Dijkstra algorithm
5. Traffic is very less	5. Traffic is very high

6. Convergence is very low	6. Convergence is faster
7. Count to infinity Problem	7. No problem of count to infinity
8. Persistent Loops	8. Transient Loops
9. RIP	9. OSPF

The maximum Hop count allowed For RIP is 15 and Hop count of 16 is considered as Destination unreachable.

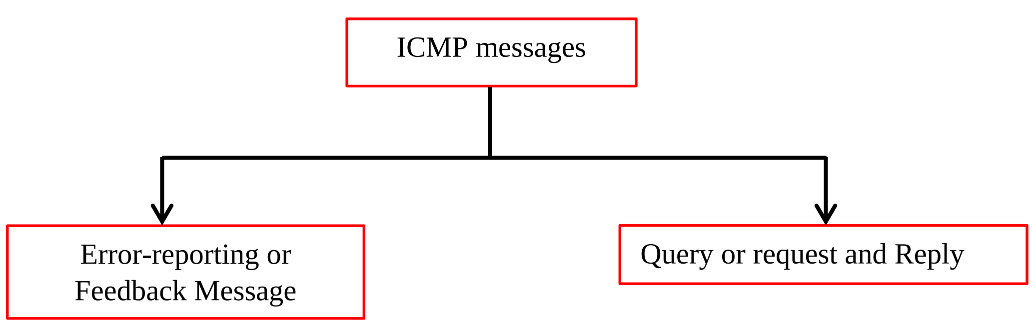
Note:
RIP uses UDP as its transport protocol with the port number – 530

7.4 IP Support Protocol

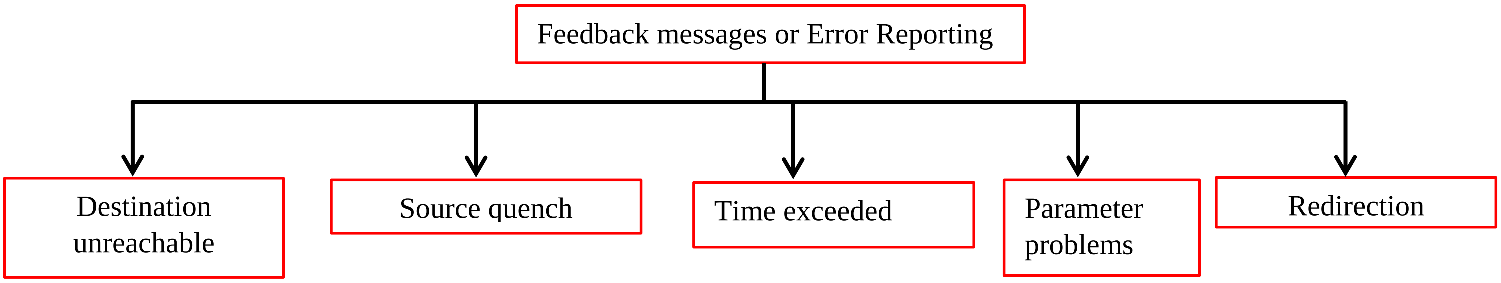
7.4.1 ARP

1. Address Resolution Protocol(ARP) is used to find the MAC(Media Access Control) address of a device from its IP address.
2. **ARP request:** ARP request is broadcasting
3. **ARP response/reply:** ARP reply is unicasting.

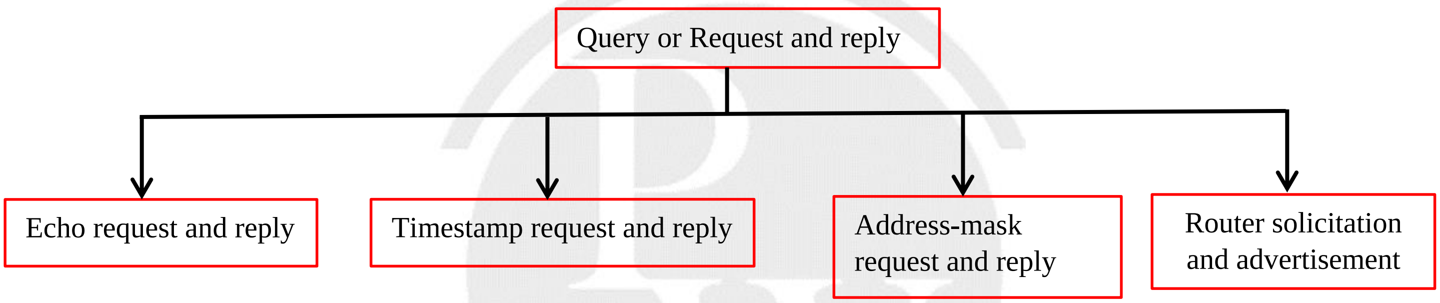
7.4.2 Internet Control Message Protocol (ICMP)



7.4.3 Internet Control Message Protocol (ICMP)



7.4.4 Internet Control Message Protocol (ICMP)



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